Computer Project 1: Drinking Water Disinfection

Background

The design of a drinking water disinfection process requires a careful balance between adding enough disinfectant to control microorganisms, without overdosing and forming unacceptably high concentrations of disinfection by-products (DBPs). This balance is affected by many factors, including disinfectant type, contact time available, temperature, pH, and dissolved organic carbon concentration. It is important that drinking water engineers and professionals understand some of the trade-offs between microbial control and DBP formation, and how to approach a treatment design to allow both objectives to be simultaneously met.

This project uses software called WatPRO that simulates the disinfection process. In particular, it calculates the *Giardia* and virus removal and inactivation credits that are achieved by a user-defined treatment train. The control of these organisms is regulated in the United States for facilities treating surface waters. WatPRO further predicts the formation of trihalomethanes, haloacetic acids, and chlorite, DBPs whose concentrations are limited by regulations due to suspected toxicity. It is the engineer's job to design a treatment train that provides adequate *Giardia* and virus control without exceeding DBP limits.

Objective

You will use WatPRO to optimize the disinfection process at a treatment plant. The treatment objectives are to meet *Giardia* and virus reduction targets without exceeding disinfection by-product (DBP) limits (see the "Procedures"). You will write a report justifying your proposed improvements to the treatment train.

Procedures

STEP 1. Input water treatment units.

The default treatment train is shown in Figure 1.

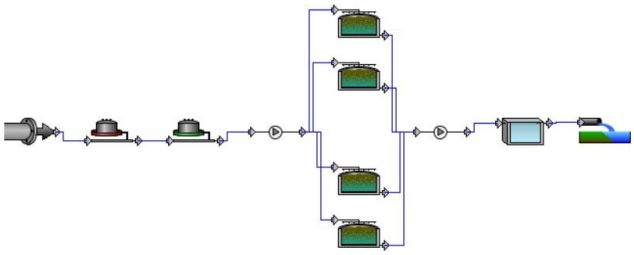


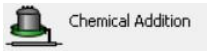
Figure 1. Water treatment plant for direct filtration.

Using the left menu, select the **Raw Water Influent** icon and drag to the layout space. If the icon is not visible, select the expansion arrow on the left to show all available unit options, then select the ***Influent*** option. This will expand the options under this menu. Look for



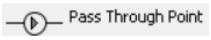
and hover the mouse cursor over this icon. Press and hold the left cursor to drag the icon to the layout space on the right.

Using the left menu, select to expand icons in the ***Chemical*** menu option, now select the **Chemical Addition** icon and drag to the layout space. Chemicals are added to the treatment train to enhance particle removal or adjust water quality parameters. The **Chemical Addition** icon should look similar to this



Using the left menu, select to expand icons in the ***Chemical*** menu option, now select the **Disinfectant Addition** icon and drag to the layout space. Disinfectants are added to the treatment train to inactivate pathogens. The **Disinfectant Addition** icon looks similar to the **Chemical Addition** icon except for the labeling.

Using the left menu, select to expand icons in the ***Distribution*** menu option, now select the **Pass Through Point** icon and drag to the layout space. This unit allows for the different flow streams to connect. The **Pass Through Point** icon should look similar to this



Using the left menu, select to expand icons in the ***Solid Separation*** menu option, now select the **Filtration** icon and drag to the layout space. Be sure to add them in a column similar to figure 1. Repeat this step three additional times. This unit removes small settleable particles in drinking water. The **Filtration** icon should look similar to this

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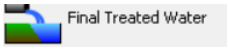
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Using the left menu, select to expand icons in the ***Distribution*** menu option, now select the **Pass Through Point** icon and drag to the layout space. Next, select to expand icons in the ***Storage*** menu option, now select the **Clear Well** icon and drag to the layout space. This unit stores treatment water ready for distribution. The **Clear Well** icon should look like this



Using the left menu, select to expand icons in the ***Effluent*** menu option, now select the **Final Treated Water** icon and drag to the layout space. The **Final Treated Water** icon should look like this



STEP 2. Connect treatment units.

Move the cursor to the connection point of the **Raw Water Influent** icon. The cursor will change from the standard white arrow ( ) to a gray directional arrow ( ). Press and hold the left-mouse clicker and drag to the inlet of connection of the **Chemical Addition** icon. The arrows will light up green when it has been connected properly. This is when you release the mouse. This connects the two units with a pipe. Repeat this process for the remaining units in the treatment sequence. For the **Pass Through Points**, connect each to the inlet of the **Filtration** icons.

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STEP 3. Set up the Site Information.

Move the cursor to the **Raw Water Influent** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. Match the influent water quality parameters to only those found in Table 1. Select *Accept* to save the changes.

Table 1. Data Entry Table for Influent

Influent Water Quality Parameter	
Flow Rate	5000 m^3/d
Temperature	12.0 deg C
pH	6.8
Total Organic Carbon	5.0 mg/L
Turbidity	20.0 NTU
Calcium Hardness (as $CaCO_3$)	80 mg/L
Total Hardness (as $CaCO_3$)	100.0 mg/L
Alkalinity (as $CaCO_3$)	100.0 mg/L
Giardia	2.0 cysts/100L

Move the cursor to the **Chemical Addition** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select Alum (Aluminum Sulfate) as the *Chemical Type* and 50.0 mg/L as the *Chemical Dosage*. Select *Accept* to save the changes.

Move the cursor to the **Disinfectant Addition** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select NaOCl – As CL2 (Anhydrous Sodium Hypochlorite) as the *Disinfectant Type* and 1.5 mg/L as the *Disinfectant Dosage*. Select *Accept* to save the changes.

Move the cursor to ONE of the **Filtration** icons. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. From the drop-down menu, select Direct as the *Filter Type*, 50.0 m^3 as the *Volume*, 30.0% as the *TOC Removal Efficiency* and 30.0% as the *UV254 Removal Efficiency*. Select *Accept* to save the changes. Repeat this for each of the four filters.

Finally, move the cursor to the **Clear Well** icon. While hovering over the icon, use the right-click button on the mouse to **Edit Parameters**. Enter 300.0 m^3 as the volume. From the drop-down menu, select Unbaffled as the *Baffling Description* and ensure that the Include in Ct(Sum) box is checked.

STEP 4. Set up the Site Information.

You may have to specify a 2nd order chlorine decay model in some versions of WatPRO. Go to the MODEL menu (located in the upper left corner of the screen), select MODEL CALIBRATION, then CHLORINE DECAY. Click on the DECAY tab, and then specify in the ORDER window, "2_{nd}".

STEP 5. Set up the Treatment Criteria.

The water quality needed for drinking water will depend on both the inactivation for pathogens and maximum levels of disinfectant (Table 2). Using the top menu, select the **Model** to pull down the options. Next select **Effluent Limits**.

In the *Data Entry* dialog box, enter the treatment criteria from Table 2. If an entry field is greyed-out, select the checkbox to the left of the target criterion. This will allow data entry. Confirm the appropriate unit of measure is applied. Unit of measures can be modified using the dropdown menu to the right of the criterion. Select *Accept* to save the changes.

Table 2. Treatment criteria

Water Quality Parameter	
Required Giardia *reduction*	10-logs
Required virus reduction	20-logs
Required Crypto reduction	4-logs
Total THM limit	30 µg/L
HAA5 limit	20 µg/L
ClO2 - limit	1 mg/L
Maximum Residual Cl2 or chloramines in effluent	2 mg/L
Maximum Residual ClO2 in effluent	0.7 mg/L
*Maximum effluent Turbidi*ty	0.5 NTU

STEP 6. Run the Model.

First complete a Connections check. Using the top menu, select the **Model** to pull down the options. Next select **Check Connections**. A pop-up window will indicate that the model is ready for analysis. Next select the **Model** to pull down the options. Now select **Calculate**. Using the **Final Effluent Summary** table at the bottom left, scroll down to check for compliance. Compliance of water quality constituents will be color-coded *green* while non-compliance will be color-coded *red*.

You can obtain more specific information by selecting various categories using the pull-down option in the **Process Output Summary** window. The options include disinfectant dose, Contact-Time Disinfectant Parameter, Pathogen, etc.

STEP 7. Conduct a Chlorine Sensitivity Analysis.

Using the top menu, select the **Analysis** to pull down the options. Next select **Sensitivity Analysis**. A new dialog box will appear. Under **Parameter Selection**, *location, parameter and effect* drop-down menus can be used to narrow the analysis parameters. For **Location** select *Disinfectant Addition*. For **Parmeter** select *Disinfectant Dosage*. This will be the variable along the graph's abscissa. For **View Effect** select *Water*. This will be the variable along the graph's ordinate.

Next, select **Range Settings**. While this is a user input variable, commonly low values are around 0.5 mg/L and high values are around 100 mg/L. The **Number of steps** is related to the mode's iterations. The higher the number (~ 500), the more iterations the model will run with increasing processing time. The lower the number (~10), the quicker the processing time, but the output will become more erratic. For this similar model, 100 iterations will be sufficient.

Then, select **Run Analysis.** A graph will appear showing the effectiveness of your treatment system at removing viruses, giardia, and cryptosporidium. For inactivation, it is typical for the colored lines to be higher than the dotted horizontal line which indicates the initial treatment criteria inputted in step 5.

Now, select **DBPs-graph** tab located on the left under **Run Analysis**. This graph shows your DBP formation. The goal for this graph is to keep the colored lines underneath the dotted horizontal lines which was our initial treatment criteria inputted in step 5.

STEP 8. Problem Analysis.

Assume that you are a consultant hired by a municipality that is concerned with ensuring adequate control of *Giardia* and viruses, without exceeding DBP limits. The municipality informs you that they would consider making the following adjustments to their present treatment train:

□ Disinfectant dose and point(s) of application

□ The use of chloramines or chlorine dioxide instead of, or in addition to, chlorine

□ The use of a contact chamber (200 m^3 maximum)

□ Building coagulation/flocculation/sedimentation units or other units

□ Increasing the alum dosage up to 50 mg/L

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Compare different alternatives and suggest the optimum treatment train to meet the municipality's concerns about microbial control and DBP concentrations given the constraints with which you are presented. Submit your findings in the format of a brief presentation. Remember to use the computer model in a realistic manner, interpreting the output using your professional judgment. If the results look wrong, explain why. Don't assume the model is necessarily correct; it's only a prediction model based on inputs.